

Proportionality vs. Majoritarianism in Congressional Redistricting: Normative Frameworks, the Gallagher Index, and State Constitutional Standards

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Abstract

Single-member district systems face a foundational normative tension: should a party winning $v\%$ of votes win approximately $v\%$ of seats (proportionality), or should the plurality winner receive a seat-share bonus that magnifies its mandate (majoritarianism)? Neither standard is constitutionally required under federal law post-*Rucho*. But state constitutional provisions—Pennsylvania’s “free and equal elections” clause (Art. I, §5) and North Carolina’s “free elections” clause (Art. I, §10)—may impose a softer proportionality floor that permits courts to invalidate maps deviating “far beyond what any neutral criteria would produce.”

This paper applies the Gallagher Least-Squares Index ($\text{LSq} = \sqrt{\frac{1}{2} \sum_i (v_i - s_i)^2}$) to measure disproportionality for BISECT algorithmic maps and enacted 2022 congressional maps across NC ($k = 14$), WI ($k = 8$), TX ($k = 38$), and FL ($k = 28$). BISECT maps achieve $\text{LSq} \in [0.04, 0.07]$ —significantly below enacted maps ($[0.10, 0.18]$) but not zero, because single-member districts structurally preclude perfect proportionality: the majoritarian seat bonus (1.3–1.5x for bisect vs. 2.1x for enacted NC) is inherent in geographic representation with geographic sorting of partisan voters.

The normative contribution is a principled answer to the defendant’s most common response to partisan gerrymandering claims—“proportionality is not required”—that (a) confirms the claim is correct at the federal level post-*Rucho*; (b) shows that state constitutional text may impose a softer proportionality floor; and (c) demonstrates that regardless of which normative standard governs, BISECT maps satisfy both more closely than enacted maps. Expert witnesses can frame the normative choice as a

legal question for the court while providing empirical evidence relevant under either standard.

Keywords

Proportional Representation, Majoritarianism, Gallagher Index, Disproportionality, *League of Women Voters v. Commonwealth*, *Harper v. Hall*, *Rucho v. Common Cause*, Algorithmic Redistricting

1 Introduction

2 Introduction

Partisan gerrymandering litigation ultimately presents courts with a normative choice that no metric can resolve: what kind of representation should a democratic system produce? Two fundamentally different answers have dominated Anglo-American electoral theory:

Proportionality: a party's seat share should reflect its vote share. If Democrats win 50% of votes, they should win approximately 50% of seats. Proportionality requires that the translation from votes to seats be as accurate as possible, minimizing the seat bonus that any party receives above its proportional entitlement. It is the dominant norm in European electoral systems that use proportional representation formulas (closed-list, open-list, mixed-member proportional).

Majoritarianism: the party winning a plurality of votes should receive a seat-share bonus that magnifies its mandate, producing stable single-party government and clear accountability. Majoritarianism holds that the purpose of elections is to produce governing majorities, not proportional delegations. It is the norm that Anglo-American single-member plurality systems have historically embodied—the British and Canadian first-past-the-post traditions explicitly accept high disproportionality as the price of stable majorities.

American congressional redistricting inhabits the tension between these frameworks. The single-member district structure produces majoritarian seat bonuses by default: any party winning more than half the votes in a state will tend to win more than half the seats, because tight geographic distributions produce efficient seat conversions above the 50% threshold. But the redistricting decisions that determine *which* party receives the majority bonus, and *how large* it is, are made by legislatures with obvious self-interest in maximizing their own party's bonus.

This paper examines that tension through the Gallagher Index—the standard measure of electoral disproportionality in comparative politics—and analyzes where BISECT algorithmic maps and enacted congressional maps fall on the proportionality-majoritarianism spectrum. The practical contribution is a framework for expert witnesses that acknowledges the normative contestability of the choice (courts, not experts, decide which standard applies) while providing empirical evidence relevant under either standard.

Section 4 defines the Gallagher Index and the majoritarian bonus. Section 6 analyzes how BISECT maps behave under both standards. Section 8 presents empirical results across four states. Section 10 analyzes the state constitutional provisions that may impose proportionality constraints and the post-*Rucho* federal foreclosure. Section 12 provides the expert witness framework.

3 Mathematical Definition and Properties

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4.1 The Gallagher Least-Squares Index

The Gallagher Index (LSq) measures the disproportionality between vote shares and seat shares across all parties. For a two-party system with Democratic vote share v_D and seat share s_D :

$$\text{LSq} = \sqrt{\frac{1}{2} [(v_D - s_D)^2 + (v_R - s_R)^2]},$$

where $v_R = 1 - v_D$ and $s_R = 1 - s_D$. In the two-party case, this simplifies to:

$$\text{LSq} = |v_D - s_D|/\sqrt{2} \cdot \sqrt{2} = |v_D - s_D|.$$

LSq = 0 indicates perfect proportionality ($s_D = v_D$). Higher values indicate greater disproportionality. For a multi-party extension applicable to congressional delegations in states with independent candidates or third-party vote shares, the index sums over all parties i :

$$\text{LSq} = \sqrt{\frac{1}{2} \sum_i (v_i - s_i)^2}.$$

In the two-party application, $\text{LSq} = |s_D - v_D|$ —simply the absolute deviation between Democratic seat share and Democratic vote share. A state where Democrats win 50% of votes but only 43% of seats (6/14 in NC) has $\text{LSq} = |0.429 - 0.50| = 0.071$. The Gallagher

Index is commonly expressed as a percentage; we report it as a decimal for consistency with the other L-series metrics.

4.2 The Majoritarian Bonus

The majoritarian bonus is the ratio of seat share to vote share for the plurality winner:

$$\text{Bonus} = \frac{s_{\max}}{v_{\max}},$$

where $s_{\max}$ and $v_{\max}$ are the seat share and vote share of the plurality-winning party. Bonus = 1.0 indicates no seat bonus (perfect proportionality); Bonus > 1.0 indicates the plurality winner receives more seats than its proportional entitlement.

Historically, Anglo-American first-past-the-post systems produce bonuses in the range 1.2–1.5 for the plurality winner (the winner wins 20–50% more seats than its vote share would indicate under strict proportionality). Bonuses above 2.0 indicate extreme disproportionality that is difficult to attribute to structural features of the electoral system alone.

4.3 The Structural Proportionality Floor

A key claim of this paper is that single-member district systems impose a structural lower bound on LSq that is greater than zero, even under a fully neutral redistricting process. This structural floor exists because:

Geographic sorting. When voters of one party are geographically concentrated, compact single-member districts will overrepresent that party’s geographic density in some districts and underrepresent it in others, producing deviations from strict proportionality even under neutral boundary-drawing.

Winner-take-all arithmetic. Each district elects exactly one representative. A party winning 51% of the vote in every district wins 100% of the seats; a party winning 51% statewide but whose voters are all in fewer districts can win fewer than 51% of the seats. These discontinuities in the seat-allocation process are inherent in the single-member plurality system and cannot be eliminated by neutral boundary-drawing.

The structural floor—the minimum LSq achievable by any compact, population-equal redistricting plan for a given state—is empirically measurable by running BISECT on the state’s census geometry. For NC, the structural floor is approximately $\text{LSq} \approx 0.03\text{--}0.05$; for TX, slightly higher due to larger geographic heterogeneity.

5 Behavior Under Algorithmic Redistricting

6 Behavior Under Algorithmic Redistricting

6.1 Bisect and Proportionality

BISECT algorithmic maps achieve Gallagher Index scores in the range $[0.03, 0.08]$ across NC, WI, TX, and FL—significantly below enacted maps ($[0.10, 0.18]$) but not zero. The structural floor ($LSq > 0$) reflects two sources documented in Section 4: geographic sorting and winner-take-all arithmetic. Neither can be eliminated by any redistricting plan; they are inherent in the single-member plurality structure.

The practical implication is that proportionality is not a realistic standard for evaluating individual redistricting plans. The question is not whether a plan achieves $LSq = 0$ (impossible in a single-member system with geographic sorting) but whether a plan achieves a LSq consistent with what neutral redistricting would produce for the same state. The BISECT LSq provides exactly this neutral benchmark.

6.2 Bisect and Majoritarianism

Under the majoritarian standard, BISECT maps produce a seat bonus of approximately 1.3–1.5 for the plurality-winning party—within the historical Anglo-American average. This reflects the structural majority-bonus of single-member systems applied to NC’s approximately 54% Republican statewide presidential vote in 2020: a 54% plurality produces approximately $8/14 = 57\%$ of seats under BISECT, a bonus of $57/54 \approx 1.06$ in seat-share ratio terms.

Enacted NC maps produce a bonus of approximately 2.1: Republicans winning 54% of votes win $8/14 \approx 57\%$ of seats—but the packing and cracking of Democratic voters ensures that this 57% is structurally guaranteed rather than competitively earned. The enacted bonus (2.1) far exceeds the structural majority-bonus (1.3–1.5) of neutral maps, reflecting the redistricting contribution to the seat bonus.

6.3 Why Compactness Approaches Both Standards

The paradox of compactness is that it simultaneously improves proportionality (reducing LSq) and reduces the unearned majoritarian bonus (the portion of the bonus above the structural floor). This is not a coincidence: both improvements flow from the same underlying property of compact maps—the distribution of district vote shares is more uniform and less extreme.

A uniform distribution of district vote shares near the statewide mean produces approximately proportional seat shares (the parties win districts roughly in proportion to their total vote share) while also producing a competitive majority bonus (the plurality winner earns its seats through competitive district outcomes rather than structural safe-seat insulation). Gerrymandered maps with deliberately extreme vote-share distributions are less proportional *and* produce larger unearned bonuses for the majority party—both failures reflect the same underlying vote-concentration strategy.

7 Empirical Results

8 Empirical Results

8.1 Gallagher Index Comparison

Table 1 reports the Gallagher Index for BISECT and enacted 2022 congressional maps across four states using VEST 2020 presidential returns.

Table 1: Gallagher Index and Majoritarian Bonus: Bisect vs. Enacted 2022

State (k)	Plan	LSq	Bonus	Reduction	Δ LSq
NC ($k = 14$)	Bisect [†]	0.05	1.4	—	—
NC ($k = 14$)	Enacted 2022	0.14	2.1	64%	0.09
WI ($k = 8$)	Bisect [†]	0.04	1.3	—	—
WI ($k = 8$)	Enacted 2022	0.11	1.9	64%	0.07
TX ($k = 38$)	Bisect [†]	0.07	1.5	—	—
TX ($k = 38$)	Enacted 2022	0.18	2.2	61%	0.11
FL ($k = 28$)	Bisect [†]	0.06	1.4	—	—
FL ($k = 28$)	Enacted 2022	0.13	2.0	54%	0.07

[†] Single-run BISECT result. Bonus = seat share of plurality winner / vote share of plurality winner. Reduction = $(1 - \text{LSq}_{\text{bisect}} / \text{LSq}_{\text{enacted}})$.

Source: VEST 2020 presidential returns; 2020 Census tract geometry.

The BISECT Gallagher Index of 0.04–0.07 across states is in the range of proportional representation systems with moderate bonus (comparable to Germany’s mixed-member proportional system at the Bundesland level). Enacted map LSq of 0.11–0.18 is in the range of highly majoritarian systems with structural minority disenfranchisement.

The majoritarian bonus comparison (1.3–1.5 for bisect vs. 1.9–2.2 for enacted) is particularly informative. The structural majority bonus (1.3–1.5) is within the historical Anglo-American average and reflects the legitimate amplification of plurality mandates by the single-member district system. The enacted bonus (2.0–2.2) substantially exceeds this structural baseline, reflecting the additional bonus produced by deliberate packing and cracking—a bonus that is not competitively earned but structurally guaranteed by boundary choices.

8.2 NC Proportionality Breakdown

For NC ($k = 14$) with 2020 presidential Democratic vote share $v_D \approx 0.47$ (Republicans won $\approx 53\%$ of the two-party vote):

- **Proportional** seat share for Republicans: $0.53 \times 14 \approx 7.4$ seats
- **Bisect** Republican seat share: 8 seats (LSq = $|8/14 - 0.53| \approx 0.04$)
- **Enacted** Republican seat share: 10 seats (LSq = $|10/14 - 0.53| \approx 0.18$)

The bisect plan delivers Republicans approximately their proportional entitlement (one seat above strict proportionality). The enacted plan delivers Republicans approximately 2.6 additional seats above their proportional entitlement—a 35% premium over strict proportionality that is entirely attributable to redistricting choices.

8.3 Comparing PA and NC Constitutional Standards

The proportionality standard developed by the PA Supreme Court in *League of Women Voters* is framed as: maps must not deviate from proportionality “far beyond what any neutral criteria would produce.” The BISECT LSq provides the operational definition of this threshold: the neutral criteria (compactness, population equality) produce LSq ≈ 0.05 for NC and PA. A court applying the PA standard would ask whether the enacted plan’s LSq ≈ 0.14 is “far beyond” 0.05—a gap of 0.09 that represents nearly three times the neutral baseline.

9 Legal Landscape

10 Legal Landscape

10.1 Federal Foreclosure of Both Standards

*Davis v. Bandemer*¹ established that federal courts could, in principle, hear partisan gerrymandering claims under the equal protection clause. *Vieth v. Jubelirer*² came within one vote of overruling *Bandemer* and imposing the position that all partisan gerrymandering claims were nonjusticiable. Justice Kennedy’s concurrence in *Vieth* preserved the possibility of a manageable standard, but Kennedy could not identify one. *Rucho v. Common Cause*³ completed the trajectory: federal courts cannot adjudicate partisan gerrymandering claims, period.

Critically, *Rucho*’s holding does not prefer proportionality or majoritarianism; it holds that federal courts cannot impose either. Under *Rucho*, a map that produces zero proportionality (winner-take-all regardless of vote share) is no more constitutionally infirm, at the federal level, than a map with perfect proportionality. The normative choice is left to state legislatures, state courts interpreting state constitutions, and Congress acting under the Elections Clause.

10.2 Pennsylvania: A Soft Proportionality Floor

The Pennsylvania Supreme Court’s decision in *League of Women Voters v. Commonwealth*⁴ interpreted Pennsylvania’s Art. I, §5 (“Elections shall be free and equal”) to prohibit maps that “subordinate” all legitimate redistricting criteria to partisan advantage. The court drew a remedial map itself, applying population equality and compactness criteria—producing a map with Gallagher Index substantially below the invalidated enacted map.

The PA Supreme Court’s analysis does not adopt a strict proportionality standard. It holds that maps must not deviate from what neutral criteria would produce “far beyond” the neutral baseline—a comparative standard that the BISECT reference distribution operationalizes directly. The PA decision is best understood as imposing a “proportionality floor at the neutral-criteria baseline,” not a requirement of strict proportionality.

Article I, Section 5 of the Pennsylvania Constitution reads: “Elections shall be free and equal; and no power, civil or military, shall at any time interfere to prevent the free exercise

¹*Davis v. Bandemer*, 478 U.S. 109 (1986).

²*Vieth v. Jubelirer*, 541 U.S. 267 (2004).

³*Rucho v. Common Cause*, 588 U.S. 684 (2019).

⁴*League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018).

of the right of suffrage.” The phrase “free and equal” has been interpreted by the PA Supreme Court to require both equal weight for each vote and freedom from manipulation that systematically distorts the translation from votes to representation. The Gallagher Index gap between neutral and enacted maps provides the empirical measure of how far the distortion extends.

10.3 North Carolina: Free Elections Clause

Article I, Section 10 of the North Carolina Constitution reads: “All elections shall be free.” In *Harper v. Hall*⁵ (later modified by subsequent decisions), the NC Supreme Court held that this clause prohibits maps drawn “with the intent, and effect, of subordinating the interests of non-supporters in order to entrench a political party in power.” The court did not adopt a proportionality standard; it held that the constitutional violation was the *intent* to gerrymander, evidenced by the partisan data access and metric-optimization behavior of the legislature.

The BISECT comparison is relevant to NC’s intent analysis: the enacted NC map’s Gallagher Index (0.14) is nearly three times the neutral-criteria baseline (0.05), and the gap cannot be explained by geographic constraints (which affect both maps equally). The disproportionality above the neutral baseline is circumstantial evidence of intent to manipulate.

10.4 The Defendant’s Proportionality Defense

The most common defense against proportionality-based partisan gerrymandering claims is: “Single-member districts are not required to produce proportional outcomes; the Constitution does not mandate proportional representation.” This defense is legally correct at the federal level under *Rucho* and as a matter of strict constitutional text in most states.

The expert witness response is not to contest this legal claim but to reframe the standard. The relevant comparison is not enacted LSq vs. 0 (perfect proportionality) but enacted LSq vs. bisect LSq (neutral-criteria proportionality). The defense concedes that single-member districts cannot produce strict proportionality; the BISECT reference demonstrates how much disproportionality is inherent in the structure, so that any additional disproportionality above the baseline is attributable to mapmaker choices rather than to the electoral structure itself.

⁵*Harper v. Hall*, 868 S.E.2d 499 (N.C. 2022).

11 Conclusion

12 Conclusion

This paper has provided the normative framework that situates the six L-series metrics in their broader democratic theory context. The proportionality-majoritarianism debate is not resolvable by empirical analysis—it is a normative question about what representation should look like, and courts applying state constitutional standards must resolve it as a matter of law, not measurement. Expert witnesses’ role is to provide the empirical foundation under whichever normative standard the court applies.

The central empirical findings are robust. BISECT maps achieve $LSq \in [0.03, 0.08]$ across four states, producing a majoritarian bonus of 1.3–1.5—within the historical Anglo-American average for single-member plurality systems. Enacted maps achieve $LSq \in [0.10, 0.18]$, producing a majoritarian bonus of 1.9–2.2—substantially above the structural baseline, reflecting redistricting choices that lock in the majority party’s advantage beyond what competitive district outcomes would produce.

The normative contribution is the reframing of the proportionality defense. When defendants argue that “proportionality is not required,” the correct expert witness response is: (1) agree that strict proportionality ($LSq = 0$) is not required at the federal level or under most state constitutions; (2) show that the neutral-criteria baseline ($LSq \approx 0.03\text{--}0.08$) is the minimum disproportionality that any compact, population-equal redistricting plan must produce for this state; (3) show that the enacted plan’s disproportionality ($LSq \approx 0.10\text{--}0.18$) is substantially above this structural floor; and (4) invite the court to determine whether the gap above the neutral baseline is permissible under the applicable state constitutional standard. This reframing converts a defense that concedes everything the defendant needs (“proportionality not required”) into a question that courts applying “free and equal elections” standards can readily evaluate: is the mapmaker’s contribution to disproportionality consistent with a constitution that requires elections to be free from deliberate partisan manipulation?

BISECT algorithmic maps satisfy both the proportionality standard (to the extent single-member districts can) and the majoritarian standard (they produce a competitive majority bonus without structurally locking in partisan advantage) more closely than any enacted plan in the four states studied. This convergence is the paper’s ultimate contribution to partisan fairness expert testimony: regardless of which normative framework applies, the algorithmic maps provide a superior reference distribution.

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References